

DD: OSR-INSTANCE-01 — Equipment Instance Registry

Developer implementation guide:

DD_OSR-INSTANCE-01-Equipment_Instance_Registry-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract

DD_OSR-INSTANCE-01-Equipment_Instance_Registry-v1.0.md.

Verdict: ■ New | **Wave:** 2 | **Group:** OSR (Operator Stock Registry) | **Version:** v1.0 | **Module:** Equipment Instance Registry

The system of record for **one row per physical, serialized device** the operator owns or has dispatched: ONTs, cable modems, MTAs, STBs, smart cards, routers. Tracks the device's identity (SKU + serial number + MAC addresses), its current location (which warehouse, which contractor van, or with which customer), and the append-only lifecycle of how it got there.

Answers the question **"where is serial number XYZ123 right now, and how did it get there?"** Owned at operator scope.

Referenced by OSR-RMA-01 (swap workflows), OSR-01 Stock Chain Engine (existing — extends SKU+quantity tracking with instance resolution for serialized SKUs), WO module (instance binding at WO finalization), PLM-CFG-06

(equipment_sku.is_serialized=TRUE SKUs MUST have rows in this table). v1.0 scope: instance master + lifecycle event ledger + current-state derivation + customer binding. Foundation for OSR-RMA-01 and any future serial-number-driven workflow.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/08_osr_equipment/DD_OSR-INSTANCE-01-Equipment_Instance_Registry-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- GET /api/equipment-instance-lifecycle-events?contractorId=wik-internal-tech-team&reasonCode=CONTRACTOR_RECOVERED_FROM_FIELD&since=2026-01-01

- GET /api/equipment-instance-lifecycle-events?originRefKind=wo&originRefId=wo_01HZ_WANJIKU_INSTALL
- GET /api/equipment-instances/{instance_id}
- GET /api/equipment-instances/{instance_id}/lifecycle-events
- GET /api/equipment-instances/{instance_id}/lifecycle-events?reasonCode=CONTRACTOR_RECOVERED_FROM_FIELD
- GET /api/equipment-instances?operator=WIK&accountId=acct_01HZ_WANJIKU
- GET /api/equipment-instances?operator=WIK&locationId=WIK-VAN-contractor_ke_nairobi_a&state=IN_CONTRACTOR_STOCK
- GET /api/equipment-instances?operator=WIK&macAddress=00:11:22:33:44:55
- GET /api/equipment-instances?operator=WIK&serial=48575443ABCD1234
- GET /api/equipment-instances?operator=WIK&skuId=WIK-ONT-HUAWEI-EG8145V5&state=IN_MAIN_WAREHOUSE
- GET /api/equipment-instances?operator=WIK&smartCard=7676543210
- GET /api/equipment-instances?serial=...
- GET /api/equipment-instances?state=IN_FIELD_ACTIVE
- PATCH /api/equipment-instances/{instance_id}
- POST /api/equipment-instances
- POST /api/equipment-instances/bulk-import
- POST /api/equipment-instances/{id}/lifecycle-events
- POST /api/equipment-instances/{instance_id}/lifecycle-events

4.2 Tables

- equipment_instance
- equipment_instance_lifecycle_event

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- sophix-superadmin
- wik-dispatcher-02
- wik-internal-tech-team
- wik-internal-tech-team-dispatcher
- wik-procurement-01
- wik-warehouse-01

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- EquipmentInstanceBoundToCustomer
- EquipmentInstanceCreated
- EquipmentInstanceDecommissioned
- EquipmentInstanceLifecycleTransition
- EquipmentInstanceRecoveredByContractor
- EquipmentInstanceUnboundFromCustomer

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

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Answers the question *"where is serial number XYZ123 right now, and how did it get there?"*

Owned at operator scope. Referenced by OSR-RMA-01 (swap workflows), OSR-01 Stock Chain Engine (existing — extends SKU+quantity tracking with instance resolution for serialized SKUs), WO module (instance binding at WO finalization), PLM-CFG-06 (`equipment_sku.is_serialized=TRUE` SKUs MUST have rows in this table).

v1.0 scope: instance master + lifecycle event ledger + current-state derivation + customer binding. Foundation for OSR-RMA-01 and any future serial-number-driven workflow.

1. What this module is

When PLM-CFG-06 flags an SKU as `is_serialized=TRUE` (every ONT, every modem, every STB, every smart card), each physical unit gets its own row in `equipment_instance`. The row carries the device's identity (serial number, MAC addresses for hybrid chassis, smart card number for STBs) and a pointer to its current location.

Location is a discriminated union: a `stock_location` from OSR-01 (warehouse or contractor van) OR an `account_id` from ILM-CFG-01 (the device is in the field at a customer premises) OR a special sentinel for `IN_TRANSIT_TO_VENDOR / DECOMMISSIONED`. Exactly one current location per active instance.

The lifecycle is captured by an append-only event ledger. Every state transition (received from vendor → moved to contractor van → assigned to WO → installed at customer → recovered after fault → returned to contractor warehouse) is one row. The current location and current state of any instance are derived from the latest event row, but the events themselves are immutable and form the audit trail.

One critical operational pain this module enables fixing: Wananchi KE reports ~40+ stale equipment items where unassigned devices on WO finalization auto-route to the main warehouse instead of the contractor team's warehouse that physically recovered them. The instance ledger captures *which contractor was holding the device when it was recovered* and downstream consumers (OSR-RMA-01, OSR-01 stock movements) use that to route the device back to the contractor team, not main warehouse.

What this module does

- Master per serialized physical device** — one row per ONT, modem, MTA, STB, smart card. Identity carries serial number, optional MAC addresses (CM-MAC, MTA-MAC, CPMAC for static IP packages, all derivable from chassis ID per workshop), smart card number for STBs (10 digits, starts with 767 for HFC TV), and the SKU FK to PLM-CFG-06.
- Track current location** — pointer to either `stock_location` (warehouse or van) OR `account` (with customer) OR a sentinel state. Updated atomically with each lifecycle event.
- Track current state** — derived enum: `IN_VENDOR_WAREHOUSE / IN_MAIN_WAREHOUSE / IN_CONTRACTOR_STOCK / RESERVED_FOR_WO / IN_FIELD_ACTIVE / IN_FIELD_DEFECTIVE / RECOVERED_BY_CONTRACTOR / RETURNED_TO_MAIN_WAREHOUSE / RMA_SHIPPED_TO_VENDOR / REPAIRED_BY_VENDOR / DECOMMISSIONED`.
- Maintain append-only lifecycle ledger** — every transition is one row with timestamp, actor, from-state, to-state, from-location, to-location, reason code, and free-text note.
- Bind / unbind to customer** — at WO finalization, the instance binds to an `account_id` (and its `subscription_id` if relevant). On disconnect / swap, the instance unbinds. The binding is the structural source of truth for *"which device does this customer have?"*
- Resolve from various lookup keys** — by serial number, by MAC address (any of CM-MAC / MTA-MAC / CPMAC), by smart card number, by SKU, by current location, by current customer.
- Bulk import** — XLSX/CSV upload (matching Wananchi's existing Equipment Batch Actions Utility) for new vendor batches. Each row creates one instance with `state = IN_MAIN_WAREHOUSE` (or `IN_VENDOR_WAREHOUSE` if marked as pre-delivery).

What this module does NOT do

- **Doesn't track non-serialized consumables.** Cable lengths, splices, drop cables, faceplates — those live in OSR-01 as SKU+quantity rows (PLM-CFG-06 `is_serialized=FALSE`). This module is exclusively for SKUs flagged `is_serialized=TRUE`.
- **Doesn't replace OSR-01.** OSR-01 still owns SKU+quantity rollups, stock movements at the SKU level, balance views, reservations. This module overlays per-instance resolution onto OSR-01's stock world for serialized items. The two coexist; OSR-01 emits a `StockMovementRecorded` event for a serialized SKU and this module captures the

corresponding EquipmentInstanceLifecycleTransition row(s) — both are written by the same business transaction.

- **Doesn't orchestrate swap workflows.** The workflows (HFC field pickup, GPON remote unassign, EQP termination retrieval, EQU upgrade) belong to OSR-RMA-01. This module only records lifecycle transitions when consumers call its APIs.
- **Doesn't talk to network systems.** Broadhub, NMS, CMTS, Verimatrix, VoipSwitch — those are downstream of provisioning workflows through capability-named provisioning adapters. This module is the BSS-side ledger of where devices are; network-side configuration is performed by Service Activation (FUL-03 → Network Provisioning Adapter) and recorded as device state by other modules.
- **Doesn't enforce slot/port binding semantics.** UC-PR4 (HFC modem authorization bound to slot/port) is a workflow-level constraint enforced by OSR-RMA-01 swap flows. This module records the slot/port as instance metadata when supplied; it doesn't validate that the slot/port matches across swaps.
- **Doesn't track warranty / repair history.** Vendor warranty terms, RMA SLA per vendor, repair history per serial — out of v1.0 scope. The lifecycle state set includes RMA_SHIPPED_TO_VENDOR and REPAIRED_BY_VENDOR as transitions, but no warranty / vendor-RMA metadata. Future expansion.
- **Doesn't perform reconciliation against the network's ground truth.** Wananchi's `malogs.wananchi.com/gpon_auto_find/` portal lists what ONT serials are physically connected to the OLT. Comparing that to instance bindings (drift detection) is a separate future reconciliation workflow.

Regional flexibility

Workshop sources are KE workshops (Zuku 2026-04-27+). KE is the v1.0 seed; schema designed to accommodate WUG/WTZ/YASSN variants:

- **MAC structure varies by tech.** HFC hybrid chassis carry CM-MAC + MTA-MAC + optional CPMAC; GPON ONT carries only its ONT serial; smart cards carry only the 10-digit number. Schema uses three nullable MAC columns + optional smart card number, not a single MAC field.
- **Slot/port semantics vary.** HFC modems bind to a CMTS slot/port (UC-PR4 — KE-specific Wananchi requirement); GPON ONTs bind to an OLT Frame/Slot/Port (F/S/P) reported by the OLT auto-discovery feed. Schema carries `network_slot_ref` as a free-text + JSONB pair to accommodate both formats without forcing a single shape.
- **Vendor RMA process varies.** WIK uses different vendors than YASSN (Huawei + FiberHome for GPON, Cisco for HFC in KE; Free Sénégal-side vendors for YASSN). The lifecycle states are vendor-agnostic; vendor metadata (RMA reference number, ship date, expected return date) is in a JSONB column rather than a structured table.

Glossary

Term	Meaning
Instance	One row in <code>equipment_instance</code> representing one physical device with a unique serial number.
Lifecycle event	One row in <code>equipment_instance_lifecycle_event</code> capturing one state transition. Append-only.
Current state	The derived state of an instance, equal to the <code>to_state</code> of its latest lifecycle event.
Current location	The derived location of an instance, equal to the <code>to_location_id</code> of its latest lifecycle event (or <code>to_account_id</code> if the instance is in the field).
Binding	The relationship <code>(instance, account)</code> when the device is at a customer premises. Established by a <code>BOUND_TO_CUSTOMER</code> event, cleared by an <code>UNBOUND_FROM_CUSTOMER</code> event.
Sentinel location	A reserved location ID used when the instance is not at a <code>stock_location</code> or customer: <code>IN_TRANSIT_TO_VENDOR</code> , <code>AT_VENDOR_FOR_REPAIR</code> , <code>DECOMMISSIONED</code> . Encoded as well-known string constants.
CM-MAC / MTA-MAC / CPMAC	The three MAC addresses an HFC hybrid chassis can carry. CM-MAC = chars 4–15 of chassis Equipment ID formatted with colon every 2 chars; MTA-MAC = CM-MAC + 2; CPMAC = optional public-IP carrier for static-IP HFC packages.
Smart card number	10-digit serial starting with 767 for HFC TV STBs (Verimatrix). Carries entitlements; printed on the card itself.

2. Tables overview + relationships

2.1 Tables overview

OSR-INSTANCE-01 owns two tables:

Table	Purpose
equipment_instance	Master per serialized physical device. Identity + current location + current state.
equipment_instance_lifecycle_event	Append-only ledger of every state / location transition for every instance.

Full DDL in §3.

2.2 Entity relationships

[Missing image: file:///home/claude/diagrams/osr_instance_erd.png]

Reading the diagram:

- `equipment_instance` is the root. One row per physical serialized device.
- `equipment_sku` (PLM-CFG-06) is referenced by `instance.sku_id` — every instance carries its SKU; the SKU defines the device kind (ONT vs modem vs STB) and `is_serialized=TRUE`.
- `stock_location` (OSR-01) is referenced by `instance.current_location_id` when the device is in a warehouse or contractor van (NOT when it's at a customer premises).
- `account` (ILM-CFG-01) is referenced by `instance.current_account_id` when the device is at a customer premises.
- `equipment_instance_lifecycle_event` is one-to-many off `equipment_instance` — every transition is one row. The latest event row determines the instance's current location and state (denormalized into the instance row for fast lookup).
- External references from this module: `stock_location` (OSR-01), `account` (ILM-CFG-01), `equipment_sku` (PLM-CFG-06), `wo` (WO-01) — never owned by OSR-INSTANCE-01.

2.3 Cross-DD anchoring

- **OSR-01 Stock Chain Engine** — for serialized SKUs, every `stock_movement` row (movement of a serialized item between locations) corresponds to one lifecycle event in this module. The two writes happen in the same business transaction. OSR-01 carries the SKU+qty rollup; OSR-INSTANCE-01 carries the per-serial detail. v1.0 design pattern: when OSR-01 records a `stock_movement` against a serialized SKU, the caller **MUST** also call `POST /api/equipment-instances/{id}/lifecycle-events` with the matching serial. Application layer enforces this via a transactional outbox.
- **PLM-CFG-06 Equipment Type Catalog** — `equipment_instance.sku_id` is FK to `equipment_sku.sku_id`. The SKU's `is_serialized` flag determines whether instances exist at all. PLM-CFG-06 also defines `equipment_kind` (ONT / CABLE_MODEM / STB / etc.) and that drives which MAC / smart-card columns are populated on the instance.
- **ILM-CFG-01 Customer Service Catalog** — `equipment_instance.current_account_id` is FK to `account.account_id` when the instance is bound to a customer. Account ownership of the device follows `account.account_id` (not `customer.customer_id`): Customer is identity-only; operational concerns including equipment binding anchor at the Account.
- **WO-01 Work Order Module** — at WO finalization, the WO carries a list of `wo_equipment(wo_id, sku_id, qty, serial_number_ref)` rows (per PLM-CFG-06 §7 migration). The `serial_number_ref` is resolved to an `equipment_instance.instance_id` and a lifecycle event `BOUND_TO_CUSTOMER` is emitted with `wo_id` as the origin reference.
- **EM-02 Contractor & Staff Registry** — every lifecycle event optionally carries a `contractor_id` (FK to EM-02) recording which contractor held the device during the transition. **THIS IS THE FIX** for the Wananchi 40+ stale-equipment routing pain: when a device is recovered from the field, the recovering contractor is recorded on the event, and downstream RMA / return-to-warehouse logic in OSR-RMA-01 reads this `contractor_id` to route the device back to the right contractor warehouse instead of main.
- **OSR-RMA-01 Equipment Swap & RMA (pending)** — the primary workflow consumer. Calls `POST /api/equipment-instances/{id}/lifecycle-events` at each swap step to advance the instance state.

3. Schemas

3.1 equipment_instance

```
CREATE TABLE equipment_instance (
  instance_id          TEXT PRIMARY KEY,          -- ULID 'inst_01HZ...'
  operator_code        TEXT NOT NULL,             -- 'WIK' | 'WUG' | 'WTZ' | 'YASSN'
  sku_id               TEXT NOT NULL REFERENCES equipment_sku(sku_id),
  serial_number        TEXT NOT NULL,             -- vendor-printed serial; primary identification
  cm_mac               TEXT NULL,                 -- HFC cable modem MAC; format AA:BB:CC:DD:EE:FF
  mta_mac              TEXT NULL,                 -- HFC MTA MAC = CM-MAC + 2 (hybrid chassis only)
  cp_mac               TEXT NULL,                 -- HFC public-IP MAC (static-IP packages only)
  smart_card_number    TEXT NULL,                 -- HFC TV STB: 10 digits, starts with '767'
  manufacturer         TEXT NULL,                 -- 'Huawei' | 'FiberHome' | 'Cisco' | 'Pace' | ...
  model                TEXT NULL,                 -- vendor model code; copied from SKU at instance creation but k
  network_slot_ref     TEXT NULL,                 -- 'F1/S2/P3' for GPON; 'cmts-nrb-01:slot-3:port-12' for HFC; fr
  network_slot_metadata JSONB NULL,               -- structured fields per tech: {'frame':'1','slot':'2','port':'3
  current_state         TEXT NOT NULL,             -- denormalized from latest lifecycle event for fast lookup
  current_location_id  TEXT NULL REFERENCES stock_location(location_id), -- NULL iff in field at customer OR sentinel s
  current_account_id    TEXT NULL REFERENCES account(account_id),      -- NULL unless in field at customer
  current_sentinel     TEXT NULL,                 -- 'IN_TRANSIT_TO_VENDOR' | 'AT_VENDOR_FOR_REPAIR' | 'DECOMMISSI
  vendor_metadata      JSONB NULL,                 -- {'vendor_ref':'RMA-2026-001','shipped_date':'2026-03-15'} whe
  active               BOOLEAN NOT NULL DEFAULT TRUE, -- false when DECOMMISSIONED (kept for history)
  created_at           TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_at           TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_by           TEXT NOT NULL,
  CHECK (current_state IN (
    'IN_VENDOR_WAREHOUSE','IN_MAIN_WAREHOUSE','IN_CONTRACTOR_STOCK','RESERVED_FOR_WO',
    'IN_FIELD_ACTIVE','IN_FIELD_DEFECTIVE','RECOVERED_BY_CONTRACTOR','RETURNED_TO_MAIN_WAREHOUSE',
    'RMA_SHIPPED_TO_VENDOR','REPAIRED_BY_VENDOR','DECOMMISSIONED'
  )),
  CHECK (
    -- Exactly one of current_location_id, current_account_id, current_sentinel is set
    (CASE WHEN current_location_id IS NOT NULL THEN 1 ELSE 0 END +
     CASE WHEN current_account_id IS NOT NULL THEN 1 ELSE 0 END +
     CASE WHEN current_sentinel IS NOT NULL THEN 1 ELSE 0 END) = 1
  ),
  CHECK (
    -- Smart card number format: 10 digits, starts with 767 (per workshop 2026-04-30)
    smart_card_number IS NULL OR smart_card_number ~ '^767[0-9]{7}$'
  )
);

CREATE UNIQUE INDEX uq_instance_serial
  ON equipment_instance (operator_code, serial_number);
CREATE UNIQUE INDEX uq_instance_cm_mac
  ON equipment_instance (operator_code, cm_mac) WHERE cm_mac IS NOT NULL;
CREATE UNIQUE INDEX uq_instance_smart_card
  ON equipment_instance (operator_code, smart_card_number) WHERE smart_card_number IS NOT NULL;
CREATE INDEX idx_instance_sku
  ON equipment_instance (operator_code, sku_id, current_state);
CREATE INDEX idx_instance_location
  ON equipment_instance (current_location_id) WHERE current_location_id IS NOT NULL;
CREATE INDEX idx_instance_account
  ON equipment_instance (current_account_id) WHERE current_account_id IS NOT NULL;
CREATE INDEX idx_instance_state
  ON equipment_instance (operator_code, current_state);
CREATE INDEX idx_instance_active_field
  ON equipment_instance (operator_code) WHERE current_state IN ('IN_FIELD_ACTIVE','IN_FIELD_DEFECTIVE');
```

Sample data:

instance_id	operator_code	sku_id	serial_number	cm_mac	mta_mac	smart_card_number	current_state	current_location_id	current_account_id
inst_01HZ_ONT_WANJIKU	WIK	WIK-ONT-HU AWEI-EG814 5V5	48575443AB CD1234	NULL	NULL	NULL	IN_FIELD_ACTIVE	NULL	acct_01HZ_WANJIKU
inst_01HZ_CM_KAMAU	WIK	WIK-CM-CIS CO-DPC3848	0011223344 556677	00:11:22:33:44:55	00:11:22:33:44:57	NULL	IN_FIELD_ACTIVE	NULL	acct_01HZ_KAMAU
inst_01HZ_STB_OTIENO	WIK	WIK-STB-PA CE-TDC575D	STB-767654 3210	NULL	NULL	7676543210	IN_FIELD_ACTIVE	NULL	acct_01HZ_OTIENO
inst_01HZ_ONT_VAN_A_01	WIK	WIK-ONT-HU AWEI-EG814 5V5	48575443BE EF5678	NULL	NULL	NULL	IN_CONTRACTOR_STOCK	WIK-VAN-contractor_ke_nairobi_a	NULL
inst_01HZ_ONT_VAN_A_02	WIK	WIK-ONT-HU AWEI-EG814 5V5	48575443CA FE9012	NULL	NULL	NULL	RESERVED_FOR_WO	WIK-VAN-contractor_ke_nairobi_a	NULL
inst_01HZ_CM_MAIN_01	WIK	WIK-CM-CIS CO-DPC3848	00AABBCCDD EEFF11	00:AA:BB:CC:DD:EE	00:AA:BB:CC:DD:F0	NULL	IN_MAIN_WAREHOUSE	WIK-CENTRAL-NRB	NULL
inst_01HZ_ONT_DEFECT_01	WIK	WIK-ONT-HU AWEI-EG814 5V5	48575443DE ADBEEF	NULL	NULL	NULL	IN_FIELD_DEFECTIVE	NULL	acct_01HZ_NJERI

instance_id	operator_code	sku_id	serial_number	cm_mac	mta_mac	smart_card_number	current_state	current_location_id	current_account_id
inst_01HZ_ONT_RECOVERED	WIK	WIK-ONT-HUAWEI-EG8145V5	48575443ABBA0001	NULL	NULL	NULL	RECOVERED_BY_CONTRACTOR	WIK-VAN-contractor_kenairobi_a	NULL
inst_01HZ_ONT_RMA_01	WIK	WIK-ONT-HUAWEI-EG8145V5	48575443BA BE2222	NULL	NULL	NULL	RMA_SHIPPED_TO_VENDOR	NULL	NULL
inst_01HZ_ONT_REPAIRED	WIK	WIK-ONT-HUAWEI-EG8145V5	48575443FA CE3333	NULL	NULL	NULL	REPAIRED_BY_VENDOR	WIK-CENTRAL-NRB	NULL
inst_01HZ_ONT_DECOMMISSIONED_01	WIK	WIK-ONT-HUAWEI-EG8145V5	48575443DE AD0000	NULL	NULL	NULL	DECOMMISSIONED	NULL	NULL
inst_01HZ_ONT_DKR_01	YASSN	YASSN-ONT-FIBERHOME-AN5506-04F	464849504B4C4D4E	NULL	NULL	NULL	IN_MAIN_WAREHOUSE	YASSN-CENTRAL-DKR	NULL

Row 1: Wanjiku has a Huawei GPON ONT installed at her premises. The ONT is in the field, bound to her account. No MAC fields populated because GPON ONTs are identified by their ONT serial alone (the OLT auto-discovers the serial; no MAC handshake at the BSS level).

Row 2: Kamau has an HFC hybrid chassis cable modem. Both CM-MAC and MTA-MAC populated; MTA-MAC = CM-MAC + 2 per workshop. No CPMAC because his package is dynamic-IP not static-IP. No smart card because this is the cable modem, not the STB.

Row 3: Otieno has an HFC TV STB. Smart card number populated (10 digits, starts with 767). No MAC fields because the STB is identified by smart card, not MAC, in the Verimatrix flow.

Rows 4–5: Two ONTs sitting in Franchise A's van. Row 4 is unreserved; row 5 has been reserved for a specific WO (state advanced to RESERVED_FOR_WO at WO finalization scheduling).

Row 6: A cable modem sitting in main warehouse, waiting to be dispatched to a contractor van.

Row 7: An ONT physically at Njeri's premises but reported defective by the customer. State = IN_FIELD_DEFECTIVE. Customer binding still active because the device hasn't been physically recovered yet. OSR-RMA-01 swap workflow will trigger a contractor visit, recover this unit, and transition it to RECOVERED_BY_CONTRACTOR.

Row 8: An ONT that was recovered from a customer's premises by Franchise A. State = RECOVERED_BY_CONTRACTOR, location = Franchise A's van. The customer binding has been cleared (NULL current_account_id). The contractor_id of the recovering team is recorded on the lifecycle event (visible in §3.2 sample data).

Row 9: An ONT shipped to vendor for repair. current_sentinel = 'IN_TRANSIT_TO_VENDOR' (deduced from current_state = 'RMA_SHIPPED_TO_VENDOR'). No location, no account, no contractor.

Row 10: A previously-shipped ONT returned from vendor, in main warehouse again. State = REPAIRED_BY_VENDOR. Can re-enter the dispatch cycle on the next outbound movement.

Row 11: An ONT that was permanently decommissioned (e.g., declared beyond repair by vendor). State = DECOMMISSIONED, active = false (not shown in this column subset).

Row 12: A YASSN-operator ONT in their central warehouse in Dakar. Different SKU vendor (FiberHome AN5506-04F vs Huawei EG8145V5 for WIK). Same schema applies.

3.2 equipment_instance_lifecycle_event

```
CREATE TABLE equipment_instance_lifecycle_event (
  event_id          TEXT PRIMARY KEY,          -- ULID 'evt_01HZ...'
  instance_id       TEXT NOT NULL REFERENCES equipment_instance(instance_id),
  event_sequence    BIGINT NOT NULL,           -- monotonically increasing per instance; the LATEST row's event_id
  from_state        TEXT NULL,                 -- NULL for the first event (instance creation)
  to_state          TEXT NOT NULL,
  from_location_id  TEXT NULL REFERENCES stock_location(location_id),
  to_location_id    TEXT NULL REFERENCES stock_location(location_id),
  from_account_id   TEXT NULL REFERENCES account(account_id),
  to_account_id     TEXT NULL REFERENCES account(account_id),
  from_sentinel     TEXT NULL,
  to_sentinel       TEXT NULL,
  reason_code       TEXT NOT NULL,             -- e.g., 'VENDOR_RECEIPT', 'TRANSFER_MAIN_TO_VAN', 'RESERVE_FOR_WO'
  contractor_id     TEXT NULL,                 -- EM-02 contractor_id of the team handling this transition. Critical
  origin_ref_kind   TEXT NULL,                 -- 'wo' | 'swap_request' | 'rma_batch' | 'manual'
```

```

origin_ref_id      TEXT NULL,                -- the wo_id, swap_request_id, rma_batch_id, or NULL for manual o
actor_user_id      TEXT NOT NULL,           -- FOUNDATION_AUTH user who performed (or system identity for aut
notes              TEXT NULL,              -- free text per workshop ('Customer reported intermittent intern
occurred_at        TIMESTAMP NOT NULL DEFAULT NOW(),
CHECK (to_state IN (
  'IN_VENDOR_WAREHOUSE', 'IN_MAIN_WAREHOUSE', 'IN_CONTRACTOR_STOCK', 'RESERVED_FOR_WO',
  'IN_FIELD_ACTIVE', 'IN_FIELD_DEFECTIVE', 'RECOVERED_BY_CONTRACTOR', 'RETURNED_TO_MAIN_WAREHOUSE',
  'RMA_SHIPPED_TO_VENDOR', 'REPAIRED_BY_VENDOR', 'DECOMMISSIONED'
)),
CHECK (origin_ref_kind IS NULL OR origin_ref_kind IN ('wo', 'swap_request', 'rma_batch', 'manual'))
);

CREATE UNIQUE INDEX uq_instance_event_seq
  ON equipment_instance_lifecycle_event (instance_id, event_sequence);
CREATE INDEX idx_instance_event_by_instance
  ON equipment_instance_lifecycle_event (instance_id, occurred_at DESC);
CREATE INDEX idx_instance_event_by_reason
  ON equipment_instance_lifecycle_event (reason_code, occurred_at DESC);
CREATE INDEX idx_instance_event_by_contractor
  ON equipment_instance_lifecycle_event (contractor_id, reason_code) WHERE contractor_id IS NOT NULL;
CREATE INDEX idx_instance_event_by_origin
  ON equipment_instance_lifecycle_event (origin_ref_kind, origin_ref_id) WHERE origin_ref_id IS NOT NULL;

```

Sample data (showing the full lifecycle of inst_01HZ_ONT_WANJIKU from Row 1 above):

event_id	instance_id	event_sequence	from_state	to_state	from_location_id	to_location_id	from_account_id	to_account_id	reason_code	contractor_id	origin_ref_kind	origin_ref_id	actor_user_id	occurred_at
evt_01_HZ_E1	inst_01HZ_ONT_WANJIKU	1	NULL	IN_VENDOR_WAREHOUSE	NULL	NULL	NULL	NULL	VENDOR_PURCHASE_ORDER_CREATED	NULL	manual	NULL	wik-procurement-01	2026-01-10T09:00:00Z
evt_01_HZ_E2	inst_01HZ_ONT_WANJIKU	2	IN_VENDOR_WAREHOUSE	IN_MAIN_WAREHOUSE	NULL	WIK-CENTRAL-NRB	NULL	NULL	VENDOR_RECEIPT	NULL	manual	NULL	wik-warehouse-01	2026-01-20T11:30:00Z
evt_01_HZ_E3	inst_01HZ_ONT_WANJIKU	3	IN_MAIN_WAREHOUSE	IN_CONTRACTOR_STOCK	WIK-CENTRAL-NRB	WIK-VAN-wik-internal-tech-team	NULL	NULL	TRANSFER_MAIN_TO_VAN	wik-internal-tech-team	manual	NULL	wik-dispatcher-02	2026-02-20T08:00:00Z
evt_01_HZ_E4	inst_01HZ_ONT_WANJIKU	4	IN_CONTRACTOR_STOCK	RESERVED_FOR_WO	WIK-VAN-wik-internal-tech-team	WIK-VAN-wik-internal-tech-team	NULL	NULL	RESERVE_FOR_WO	wik-internal-tech-team	wo	wo_01HZ_WANJIKU_IN_STALL	wik-internal-tech-team-dispatcher	2026-02-22T17:00:00Z
evt_01_HZ_E5	inst_01HZ_ONT_WANJIKU	5	RESERVED_FOR_WO	IN_FIELD_ACTIVE	WIK-VAN-wik-internal-tech-team	NULL	NULL	acct_01HZ_WANJIKU	WO_INSTALL_FINALIZED	wik-internal-tech-team	wo	wo_01HZ_WANJIKU_IN_STALL	tech_wik_internal_01	2026-02-23T15:45:00+03:00

Row 1: ONT created in the instance registry the moment the procurement team enters the purchase order. State = IN_VENDOR_WAREHOUSE; the device hasn't physically arrived yet, but the BSS knows about it. No location, no account, no contractor.

Row 2: ONT physically arrives at WIK Central Warehouse Nairobi. State transitions to IN_MAIN_WAREHOUSE, location set to WIK-CENTRAL-NRB. Reason: VENDOR_RECEIPT. Actor: warehouse staff who scanned it in.

Row 3: Two weeks later, the dispatcher transfers a batch of ONTs (including this one) from main warehouse to the WIK Internal Tech Team's van. State = IN_CONTRACTOR_STOCK, location = WIK-VAN-wik-internal-tech-team. contractor_id populated with the receiving team's ID — this is the critical field that downstream return-routing reads.

Row 4: Two days before Wanjiku's install appointment, the dispatcher reserves this specific ONT for her WO. State = RESERVED_FOR_WO, location stays at the van but it's no longer available for other WOs. origin_ref_kind='wo', origin_ref_id carries the WO ID.

Row 5: On install day, the technician finalizes the WO. State = IN_FIELD_ACTIVE, to_location_id becomes NULL (device is in the field, not at a stock location), to_account_id carries Wanjiku's account. Contractor still recorded — the WIK Internal Tech Team is the contractor who placed it at the customer's premises. This is the contractor that, if the device ever fails and needs recovery, will own the routing back to their warehouse.

The denormalized fields on `equipment_instance` (`current_state`, `current_location_id`, `current_account_id`) are kept in sync with the latest event's `to_*` fields by the application layer in the same transaction that inserts the event. The `event_sequence` provides optimistic concurrency control: a write that doesn't match `current_event_sequence + 1` aborts.

4. Rules

Rule	Description
R-OSR-INST-1	One instance per (<code>operator_code</code> , <code>serial_number</code>). The serial number is the primary natural key; two operators may share serial collisions across different vendors (handled by the <code>operator_code</code> partition in the unique index).
R-OSR-INST-2	Only SKUs with <code>is_serialized=TRUE</code> in PLM-CFG-06 may have instances. INSERT is rejected if the SKU is not serialized. Migration tooling for the OSR-01 transition handles legacy SKU+qty rollups; net-new instances always go through the typed API.
R-OSR-INST-3	An instance has exactly one current location: either <code>current_location_id</code> (warehouse / van), <code>current_account_id</code> (customer premises), or <code>current_sentinel</code> ('IN_TRANSIT_TO_VENDOR' / 'AT_VENDOR_FOR_REPAIR' / 'DECOMMISSIONED'). Enforced by CHECK constraint.
R-OSR-INST-4	Every state transition is one row in <code>equipment_instance_lifecycle_event</code> . The instance row's denormalized <code>current_*</code> fields MUST equal the latest event row's <code>to_*</code> fields after the transaction. Application layer enforces this.
R-OSR-INST-5	<code>event_sequence</code> is per-instance, monotonically increasing, no gaps. Each new event MUST have <code>event_sequence = previous_event_sequence + 1</code> . Concurrency conflict (two clients trying to write event sequence N+1 simultaneously) is detected by the UNIQUE constraint and one client receives a 409 — that client must re-read the instance and retry.
R-OSR-INST-6	Legal state transitions are enforced at the application layer per a state machine (§4.1). Illegal transitions (e.g., DECOMMISSIONED → IN_FIELD_ACTIVE) are rejected by the API with HTTP 409 + a <code>code=ILLEGAL_STATE_TRANSITION</code> body.
R-OSR-INST-7	When an instance transitions to RECOVERED_BY_CONTRACTOR, the event's <code>contractor_id</code> MUST be populated. This is the field downstream routing logic (OSR-RMA-01) reads to decide "route this device back to which warehouse?" — fixing the Wananchi 40+ stale-equipment pain. Application layer enforces via NOT NULL guard on this specific <code>reason_code</code> .
R-OSR-INST-8	Customer binding lifecycle: state IN_FIELD_ACTIVE or IN_FIELD_DEFECTIVE requires <code>current_account_id</code> set and <code>current_location_id</code> NULL. All other states require <code>current_account_id</code> NULL. Enforced by CHECK constraint + application layer.
R-OSR-INST-9	DECOMMISSIONED is terminal. Once an instance enters this state, no further lifecycle events are accepted (application layer rejects with HTTP 409 + <code>code=INSTANCE_DECOMMISSIONED</code>). The instance row's <code>active</code> field is set to false. Row is kept indefinitely for audit history.
R-OSR-INST-10	Smart card number format: must match <code>^767[0-9]{7}\$</code> (10 digits starting 767, per workshop 2026-04-30 — Verimatrix-issued smart cards for HFC TV). Enforced by CHECK constraint.
R-OSR-INST-11	MAC address format: AA:BB:CC:DD:EE:FF (colon-separated hex pairs, case-insensitive). Application layer normalizes to uppercase on write. CM-MAC and MTA-MAC values per workshop: MTA-MAC computed as CM-MAC + 2; application layer may validate this for hybrid chassis SKUs but does NOT auto-populate (the import file is the source of truth).
R-OSR-INST-12	<code>network_slot_metadata</code> JSONB shape varies by tech: for GPON <code>{'frame':int,'slot':int,'port':int,'olt':text}</code> ; for HFC <code>{'cmts':text,'slot':int,'port':int,'authz_bound':bool}</code> . Application layer parses + validates per the SKU's <code>equipment_kind</code> (PLM-CFG-06). Unknown shapes are accepted (stored as-is) for forward compatibility.
R-OSR-INST-13	Cross-operator scoping: <code>operator_code</code> on <code>equipment_instance</code> MUST match the <code>operator_code</code> of the SKU referenced (PLM-CFG-06 SKUs are operator-scoped). Cross-operator instance creation is rejected. Cross-operator query requires <code>sophix-superadmin</code> role.
R-OSR-INST-14	<code>vendor_metadata</code> JSONB is populated only when the instance is in RMA_SHIPPED_TO_VENDOR or REPAIRED_BY_VENDOR state. Cleared (set to NULL) on transition out of either state. Application layer enforces.

4.1 State machine

Legal transitions (rows = from, columns = to):

From \ To	IN_VEND OR_WH	IN_MAIN_ WH	IN_CONTRACTOR_ STOCK	RESERVED_FOR_W O	IN_FIELD_ ACTIVE	IN_FIELD_ DEFECTIVE	RECOVERED	RETURNED_TO_MA IN	RMA_SHIPPED	REPAIRED	DECOMMISSIONED
---	---	---	---	---	---	---	---	---	---	---	---
(initial creation)	✓	✓									
IN_VEND OR_WAREHOUSE		✓									✓
IN_MAIN_ WAREHOUSE			✓						✓		✓
IN_CONTRACTOR_ STOCK		✓		✓							✓
RESERVED_FOR_W O			✓		✓						✓
IN_FIELD_ ACTIVE						✓	✓				
IN_FIELD_ DEFECTIVE					✓		✓				
RECOVERED_BY_ CONTRACTOR			✓					✓			✓
RETURNED_TO_MA IN_WAREHOUSE		✓							✓		✓
RMA_SHIPPED_TO_ VENDOR										✓	✓
REPAIRED_BY_ VENDOR		✓									✓
DECOMMISSIONED	terminal	terminal	terminal	terminal	terminal	terminal	terminal	terminal	terminal	terminal	terminal

Notes on the state machine:

- IN_FIELD_DEFECTIVE → IN_FIELD_ACTIVE is allowed (a customer-reported issue resolved without truck roll — e.g., remote reboot fixed it; the swap workflow cancels).
- RECOVERED_BY_CONTRACTOR → IN_CONTRACTOR_STOCK is the swap-success path: contractor brings the recovered unit back to van; it's now part of contractor stock awaiting return-to-main or RMA decision.
- RECOVERED_BY_CONTRACTOR → RETURNED_TO_MAIN_WAREHOUSE skips the contractor stock step when the contractor batch-transfers recovered units directly to main warehouse.
- RETURNED_TO_MAIN_WAREHOUSE → IN_MAIN_WAREHOUSE moves the unit back into available stock after main warehouse inspection (a recovered unit that tests OK can be re-dispatched).
- RETURNED_TO_MAIN_WAREHOUSE → RMA_SHIPPED_TO_VENDOR is the path for units failing main-warehouse inspection.
- Most states can transition to DECOMMISSIONED (write-off allowed at almost any point — lost, stolen, declared beyond repair).

5. REST API

5.1 Instance master CRUD

```
POST /api/equipment-instances HTTP/1.1
Authorization: Bearer <service-jwt>
Content-Type: application/json
```

```
{
  "operatorCode":      "WIK",
  "skuId":             "WIK-ONT-HUAWEI-EG8145V5",
  "serialNumber":      "48575443ABCD1234",
  "manufacturer":      "Huawei",
  "model":             "EG8145V5",
  "initialState":      "IN_VENDOR_WAREHOUSE",
  "initialReasonCode": "VENDOR_PURCHASE_ORDER_CREATED",
  "actorUserId":       "wik-procurement-01",
  "notes":             "PO# WIK-PO-2026-002, 200-unit batch"
}
```

→ 201 Created

```
{
  "instanceId":      "inst_01HZ_ONT_NEW_BATCH_47",
  "operatorCode":     "WIK",
  "skuId":            "WIK-ONT-HUAWEI-EG8145V5",
  "serialNumber":     "48575443ABCD1234",
  "currentState":     "IN_VENDOR_WAREHOUSE",
  "currentLocationId": null,
  "currentAccountId": null,
  "currentSentinel":  null,
  "active":           true,
  "createdAt":        "2026-01-10T09:00:00Z"
}
```

```
GET /api/equipment-instances/{instance_id}
```

```
GET /api/equipment-instances?operator=WIK&serial=48575443ABCD1234
```

```
GET /api/equipment-instances?operator=WIK&smartCard=7676543210
```

```
GET /api/equipment-instances?operator=WIK&macAddress=00:11:22:33:44:55 # matches cm_mac, mta_mac, or cp_mac
```

```
GET /api/equipment-instances?operator=WIK&accountId=acct_01HZ_WANJIKU # all devices at a customer
```

```
GET /api/equipment-instances?operator=WIK&locationId=WIK-VAN-contractor_ke_nairobi_a&state=IN_CONTRACTOR_STOCK
```

```
GET /api/equipment-instances?operator=WIK&skuId=WIK-ONT-HUAWEI-EG8145V5&state=IN_MAIN_WAREHOUSE
```

```
PATCH /api/equipment-instances/{instance_id} # update metadata fields (model, manufact
```

5.2 Lifecycle event recording

The primary write endpoint — every state/location change goes through here:

```
POST /api/equipment-instances/{instance_id}/lifecycle-events HTTP/1.1
Authorization: Bearer <service-jwt>
Content-Type: application/json
```

```
{
  "toState":      "IN_FIELD_ACTIVE",
  "toAccountId":  "acct_01HZ_WANJIKU",
  "reasonCode":   "WO_INSTALL_FINALIZED",
  "contractorId": "wik-internal-tech-team",
  "originRefKind": "wo",
  "originRefId":  "wo_01HZ_WANJIKU_INSTALL",
  "actorUserId":  "tech_wik_internal_01",
  "notes":        "ONT installed at customer premises; OLT auto-discovered serial; final WO finalization at 15:45",
  "expectedEventSequence": 4
}
```

→ 201 Created

```
{
  "eventId":      "evt_01HZ_WANJIKU_INSTALL",
  "instanceId":   "inst_01HZ_ONT_WANJIKU",
  "eventSequence": 5,
  "fromState":    "RESERVED_FOR_WO",
  "toState":      "IN_FIELD_ACTIVE",
  "fromLocationId": "WIK-VAN-wik-internal-tech-team",
  "toLocationId":  null,
  "fromAccountId": null,
  "toAccountId":   "acct_01HZ_WANJIKU",
  "reasonCode":    "WO_INSTALL_FINALIZED",
  "contractorId":  "wik-internal-tech-team",
  "originRefKind": "wo",
  "originRefId":   "wo_01HZ_WANJIKU_INSTALL",
  "actorUserId":   "tech_wik_internal_01",
  "occurredAt":    "2026-02-23T15:45:00+03:00"
}
```

Concurrency: if expectedEventSequence doesn't equal the instance's current event_sequence, the API returns 409:

```
→ 409 Conflict
{
```

```

    "code": "STALE_EVENT_SEQUENCE",
    "message": "Expected event sequence 4 but instance is at 5. Re-fetch and retry.",
    "currentEventSequence": 5
}

```

Illegal state transition:

```

→ 409 Conflict
{
  "code": "ILLEGAL_STATE_TRANSITION",
  "message": "Cannot transition instance from DECOMMISSIONED to IN_FIELD_ACTIVE (terminal state).",
  "currentState": "DECOMMISSIONED"
}

```

5.3 Lifecycle event read

```

GET /api/equipment-instances/{instance_id}/lifecycle-events # full history, newest first
GET /api/equipment-instances/{instance_id}/lifecycle-events?reasonCode=CONTRACTOR_RECOVERED_FROM_FIELD
GET /api/equipment-instance-lifecycle-events?contractorId=wik-internal-tech-team&reasonCode=CONTRACTOR_RECOVERED_FROM_FIELD
GET /api/equipment-instance-lifecycle-events?originRefKind=wo&originRefId=wo_01HZ_WANJIKU_INSTALL # all events tied to

```

5.4 Bulk import (workshop alignment)

Mirrors the Wananchi Equipment Batch Actions Utility pattern (workshop 2026-04-30: upload book file → assign status → process):

```

POST /api/equipment-instances/bulk-import HTTP/1.1
Authorization: Bearer <bo-user-jwt>
Content-Type: application/json

{
  "operatorCode": "WIK",
  "skuId": "WIK-ONT-HUAWEI-EG8145V5",
  "initialState": "IN_MAIN_WAREHOUSE",
  "initialLocationId": "WIK-CENTRAL-NRB",
  "initialReasonCode": "VENDOR_RECEIPT",
  "actorUserId": "wik-warehouse-01",
  "instances": [
    { "serialNumber": "48575443ABCD1234", "manufacturer": "Huawei", "model": "EG8145V5" },
    { "serialNumber": "48575443ABCD1235", "manufacturer": "Huawei", "model": "EG8145V5" },
    { "serialNumber": "48575443ABCD1236", "manufacturer": "Huawei", "model": "EG8145V5" }
  ],
  "batchRef": "VENDOR-PO-2026-002-batch-A"
}

→ 200 OK
{
  "imported": 3,
  "rejected": 0,
  "rowResults": [
    { "row": 1, "serialNumber": "48575443ABCD1234", "instanceId": "inst_01HZ_BATCH_001", "status": "created" },
    { "row": 2, "serialNumber": "48575443ABCD1235", "instanceId": "inst_01HZ_BATCH_002", "status": "created" },
    { "row": 3, "serialNumber": "48575443ABCD1236", "instanceId": "inst_01HZ_BATCH_003", "status": "created" }
  ],
  "batchRef": "VENDOR-PO-2026-002-batch-A"
}

```

Partial-reject mode (default): rows with duplicate serial are reported but don't block the batch. Hard mode

(mode=all-or-nothing query param): any duplicate aborts the batch.

For HFC modems with hybrid chassis (CM-MAC + MTA-MAC + optional CPMAC), the bulk import payload row carries those fields:

```

{
  "serialNumber": "HFC-CHASSIS-707C4-A123",
  "manufacturer": "Cisco",
  "model": "DPC3848",
  "cmMac": "00:11:22:33:44:55",
  "mtaMac": "00:11:22:33:44:57",
  "cpMac": null
}

```

For STBs with smart cards:

```

{
  "serialNumber": "STB-7676543210",
  "manufacturer": "Pace",
  "model": "TDC575D",
  "smartCardNumber": "7676543210"
}

```

6. Kafka events

Topics: `sophix.osr.instance.*`

Event	Topic	Fires on	Consumers
EquipmentInstanceCreated	<code>sophix.osr.instance.created</code>	New instance row inserted	OSR-01 (sync stock_balance), EM-04 Reporting (pending)
EquipmentInstanceLifecycleTransition	<code>sophix.osr.instance.lifecycle-transition</code>	Any state transition (including location change without state change)	OSR-01 (sync stock movement for serialized SKUs) , OSR-RMA-01 (drive swap workflows), WO-01 (mark wo_equipment finalized), BIL-01 Charging Engine (filter on reason=WO_INSTALL_FINALIZED to trigger deposit charge; filter on reason=CONTRACTOR_RECOVERED_FROM_FIELD or RETURN_VAN_TO_MAIN to trigger deposit refund)
EquipmentInstanceBoundToCustomer	<code>sophix.osr.instance.bound-to-customer</code>	Transition to IN_FIELD_ACTIVE with current_account_id populated	NOT-01 (customer notification), ICN-01 (internal staff notification: "your install completed"), BIL-01 (deposit recognition), reporting
EquipmentInstanceUnboundFromCustomer	<code>sophix.osr.instance.unbound-from-customer</code>	Transition out of IN_FIELD_ACTIVE / IN_FIELD_DEFECTIVE to non-customer state	NOT-01, ICN-01, BIL-01 (deposit refund eligibility)
EquipmentInstanceRecoveredByContractor	<code>sophix.osr.instance.recovered-by-contractor</code>	Transition to RECOVERED_BY_CONTRACTOR	OSR-RMA-01 (drive routing-back logic), EM-04 Reporting
EquipmentInstanceDecommissioned	<code>sophix.osr.instance.decommissioned</code>	Transition to DECOMMISSIONED	OSR-01 (zero out the SKU's contribution to balance), BIL-01 (write-off accounting), EM-04 Reporting

6.1 Sample event — EquipmentInstanceLifecycleTransition (WO install)

```
{
  "eventType": "EquipmentInstanceLifecycleTransition",
  "aggregateId": "inst_01HZ_ONT_WANJIKU",
  "timestamp": "2026-02-23T15:45:00.000Z",
  "payload": {
    "operatorCode": "WIK",
    "instanceId": "inst_01HZ_ONT_WANJIKU",
    "skuId": "WIK-ONT-HUAWEI-EG8145V5",
    "serialNumber": "48575443ABCD1234",
    "eventSequence": 5,
    "lifecycleEventId": "evt_01HZ_WANJIKU_INSTALL",
    "fromState": "RESERVED_FOR_WO",
    "toState": "IN_FIELD_ACTIVE",
    "fromLocationId": "WIK-VAN-wik-internal-tech-team",
    "toLocationId": null,
    "fromAccountId": null,
    "toAccountId": "acct_01HZ_WANJIKU",
    "fromSentinel": null,
    "toSentinel": null,
    "reasonCode": "WO_INSTALL_FINALIZED",
    "contractorId": "wik-internal-tech-team",
    "originRefKind": "wo",
    "originRefId": "wo_01HZ_WANJIKU_INSTALL",
    "actorUserId": "tech_wik_internal_01",
    "occurredAt": "2026-02-23T15:45:00+03:00"
  }
}
```

6.2 Sample event — EquipmentInstanceRecoveredByContractor (THE Wananchi routing fix)

```
{
  "eventType": "EquipmentInstanceRecoveredByContractor",
  "aggregateId": "inst_01HZ_ONT_NJERI",
  "timestamp": "2026-03-12T11:20:00.000Z",
  "payload": {
    "operatorCode": "WIK",
    "instanceId": "inst_01HZ_ONT_NJERI",
    "skuId": "WIK-ONT-HUAWEI-EG8145V5",
    "serialNumber": "48575443DEADBEEF",
    "eventSequence": 6,
  }
}
```

```

    "lifecycleEventId": "evt_01HZ_NJERI_RECOVERED",
    "fromState": "IN_FIELD_DEFECTIVE",
    "toState": "RECOVERED_BY_CONTRACTOR",
    "fromLocationId": null,
    "toLocationId": "WIK-VAN-contractor_ke_nairobi_a",
    "fromAccountId": "acct_01HZ_NJERI",
    "toAccountId": null,
    "reasonCode": "CONTRACTOR_RECOVERED_FROM_FIELD",
    "contractorId": "contractor_ke_nairobi_a",
    "originRefKind": "swap_request",
    "originRefId": "swap_01HZ_NJERI_MODEM_FAULT",
    "actorUserId": "tech_franchise_a_03",
    "occurredAt": "2026-03-12T11:20:00+03:00",
    "notes": "Customer reported intermittent connection; tech replaced ONT on site; defective unit returned to
  }
}

```

The `contractorId` field is what OSR-RMA-01 reads to know "route the defective unit back through Franchise A's warehouse, not main." This is the explicit fix for the documented 40+ stale-equipment routing pain.

7. Cross-DD coordination

7.1 Upstream callers

Caller	Endpoint	When
Procurement (pending DD)	POST /api/equipment-instances/bulk-import	New vendor batch received at main warehouse
OSR-01 stock-movement	POST /api/equipment-instances/{id}/lifecycle-events	For serialized SKUs, every stock movement triggers an instance lifecycle event in the same transaction
WO-01 finalization	POST /api/equipment-instances/{id}/lifecycle-events	At WO finalization, the technician's reported serial(s) are bound to the customer
OSR-RMA-01 (pending)	POST /api/equipment-instances/{id}/lifecycle-events	Throughout swap workflows: customer-reported defect, contractor recovery, return-to-warehouse, RMA-ship, etc.
BO admin UI	GET /api/equipment-instances?serial=...	Operator support: "where is serial XYZ?" lookup
Network reconciliation (future DD)	GET /api/equipment-instances?state=IN_FIELD_ACTIVE	Daily drift detection vs OLT auto-discovery

7.2 Downstream consumers (Kafka)

Consumer	Events subscribed	Action
OSR-01 Stock Chain Engine	EquipmentInstanceLifecycleTransition (for serialized SKUs)	Sync stock_balance + emit corresponding stock_movement row
OSR-RMA-01 (pending)	EquipmentInstanceLifecycleTransition, EquipmentInstanceRecoveredByContractor	Drive swap workflows; route recovered units to contractor warehouse
WO-01	EquipmentInstanceBoundToCustomer, EquipmentInstanceUnboundFromCustomer	Update wo_equipment status; close WO when binding event arrives
BIL-01 Charging Engine	EquipmentInstanceBoundToCustomer (filter reason=WO_INSTALL_FINALIZED), EquipmentInstanceUnboundFromCustomer (filter reason=CONTRACTOR_RECOVERED_FROM_FIELD)	Apply equipment deposit charge at install; trigger refund adjustment at recovery
NOT-01	EquipmentInstanceBoundToCustomer	Customer-facing notification: "your install is complete; device serial XYZ activated"
ICN-01	EquipmentInstanceRecoveredByContractor	Notify contractor's main handler: "device XYZ recovered, awaiting van check-in"
EM-04 Reporting (pending)	All <code>sophix.osr.instance.*</code>	Equipment lifecycle metrics: install rates, defect rates, mean-time-to-recover, per-contractor recovery volumes

7.3 Sibling modules

Module	Direction	Contract
OSR-01 Stock Chain Engine	Bidirectional	OSR-01 owns SKU+qty rollups; OSR-INSTANCE-01 owns per-serial detail. Every serialized SKU stock_movement (OSR-01) corresponds to one or more lifecycle events (OSR-INSTANCE-01). Transactional outbox ensures consistency.
PLM-CFG-06 Equipment Type Catalog	OSR-INSTANCE-01 → PLM-CFG-06 (read)	equipment_instance.sku_id is FK to equipment_sku.sku_id. PLM-CFG-06 is upstream; no reverse FK.
ILM-CFG-01 Customer Service Catalog	OSR-INSTANCE-01 → ILM-CFG-01 (read)	equipment_instance.current_account_id is FK to account.account_id. Binding is at account level, not customer level.
EM-02 Contractor & Staff Registry	OSR-INSTANCE-01 → EM-02 (read)	equipment_instance_lifecycle_event.contractor_id is FK to contractor.contractor_id. Critical for return-routing logic.
WO-01 Work Order Module	OSR-INSTANCE-01 → WO-01 (read)	equipment_instance_lifecycle_event.origin_ref_id (when origin_ref_kind='wo') is FK to wo.wo_id.

7.4 Dependencies (build order)

OSR-INSTANCE-01 v1.0 deploys after:

1. **PLM-CFG-06** (equipment_sku table must exist for FK reference)
2. **OSR-01** (stock_location table must exist for FK reference)
3. **ILM-CFG-01** (account table must exist for FK reference)
4. **EM-02** (contractor table must exist for contractor_id reference on lifecycle events; deployable in parallel since the FK is loosely typed)
5. **WO-01** (wo table must exist for origin_ref_id when origin_ref_kind='wo'; deployable in parallel)

OSR-RMA-01 deploys after OSR-INSTANCE-01.

8. Workshop sources

Workshop sources for v1.0 (KE Zuku 2026-04-27+; no fabricated TZ baselines — TBD pending cartography):

- **Workshop 2026-04-29 (Work Order & Field Operations deep dive)** — Equipment Lookup screen behavior (search by last 5 hex digits returns all sub-interfaces of a chassis, e.g. 707C4 → CMD/MTP/MTA); Hybrid Chassis dialog (HYB chassis carries CM-MAC + MTA-MAC; CM-MAC = chars 4-15 of chassis Equipment ID, MTA-MAC = CM-MAC + 2); Equipment Batch Actions Utility on named persistent lists.
- **Workshop 2026-04-30 (Provisioning deep dive)** — Smart card 10-digit format starting 767 (Verimatrix-issued for HFC TV); HFC modem authorization bound to slot/port (UC-PR4, KE-specific — moving the modem to a different slot breaks authorization); static-IP HFC requires double config (Broadhub + Clearcable NOMS); static-IP GPON manual via DHCP console; voice device (MTA/IAD) can carry multiple phone numbers each bound to an equipment port.
- **Workshop 2026-04-30 (Customer Management deep dive — Equipment-routing gap)** — Today: unassigned equipment goes to main warehouse on WO finalization, not to the contractor team that recovered it. ~40+ stale items reported. Wananchi target: SOPHIX must trace the contractor team that recovered the equipment and route it back to that team's warehouse. THIS DD'S contractor_id on equipment_instance_lifecycle_event is the mechanism that enables that fix in OSR-RMA-01.
- **Workshop 2026-04-30 (Provisioning deep dive — Service Termination)** — Equipment dismantle is a separate retrieval WO if customer doesn't bring CPE back. For both GPON and HFC, an agent contractor is sent on site. Refund processed via Invoice Adjustments. This drives the lifecycle states IN_FIELD_DEFECTIVE and RECOVERED_BY_CONTRACTOR and the contractor_id tracking.
- **wo_lifecycle.md Job Type table** — EQP Equipment Pickup, EQU Equipment Upgrade, EQR Equipment Recovery (failed: "client paid for CPE & refuses return"). These map to OSR-RMA-01 satellite workflows; this DD's instance state machine accommodates them via DECOMMISSIONED reachability and RECOVERED_BY_CONTRACTOR semantics.

- **BP-08 Equipment Swap (RMA) GAP analysis** — Anchor Wananchi: technology-specific swap (HFC modem = field pickup, GPON ONT = remote unassign + new bind), device authorization slot/port mandatory (UC-PR4), serial unbind/rebind (UC-PR9), stock movement default main warehouse — link contractor stock broken (documented gap). G-08-1 (workflow) and G-08-2 (contractor stock traceability) are addressed by this DD + OSR-RMA-01 (pending).

TZ codebase coverage today: to be filled in after codebase review (no equivalent of per-instance lifecycle ledger has been confirmed in the TZ codebase yet; SIP-05 has SKU+qty CRUD + bulk import without per-serial lifecycle tracking, but full assessment pending cartography).

Gap & V3 design rationale: OSR-INSTANCE-01 introduces the per-instance ledger that makes the documented Wananchi contractor-stock-routing gap fixable. The schema design preserves the bulk-import pattern Wananchi already uses (Equipment Batch Actions Utility) while adding the per-event audit + contractor_id tracking that today's BSS lacks.